

AI Prompts Used During Project Development

1. Understanding the Project

Prompt 1

I got this CN project on TLS fingerprinting using JA3/JA3S. Before we start coding can you explain the whole project to me in simple terms and what exactly we are expected to build?

Prompt 2

First explain the TLS handshake to me properly. What happens when I do https requests and what exactly are ClientHello and ServerHello?

Prompt 3

Okay now explain ClientHello in more detail. What all information is present inside it and which of those things can we actually see by sniffing packets?

Prompt 4

What exactly is TLS fingerprinting? How can we identify curl or browser just by looking at ClientHello if we are not decrypting anything?

Prompt 5

Now explain JA3 from scratch. What fields does it use and how do those fields finally become one hash?

Prompt 6

Can you take one example of ClientHello and manually show me how the JA3 string is constructed and then converted to MD5?

Prompt 7

I keep seeing GREASE in JA3. What exactly is GREASE and why do we remove it while calculating JA3?

Prompt 8

Okay understood JA3. Now explain JA3S and how it is different from JA3. What information are we getting from ServerHello?

Prompt 9

How does passive packet capture actually work here? If we are just sniffing traffic, how does Scapy get the ClientHello and ServerHello without becoming a MITM?

Prompt 10

Now connect everything once. If I run curl to example.com, what exactly happens from packet capture till we finally print the JA3 hash and identify curl?

2. HLD / Design Discussion

Prompt 11

I want to keep this project simple and suitable for a CN course. Can we use Scapy for capture, our own Python code for JA3 and Redis just for storing the fingerprints? Is this enough?

Prompt 12

For ClientHello parsing, should we use Scapy's TLS fields directly or should we parse the raw bytes ourselves? Which approach makes more sense for this project?

Prompt 13

Before coding, review this architecture once and tell me if there is anything unnecessary or anything missing from the actual project requirements.

3. Designing the TLS Parser

Prompt 14

I want to parse the raw TLS handshake bytes myself instead of using a complete TLS parsing library. What fields do I need to extract for JA3 and JA3S?

Prompt 15

Explain the TLS ClientHello byte structure and show me how I can move through it using offsets.

Prompt 16

Can you help me implement parser.py which takes raw ClientHello/ServerHello bytes and returns only the fields required for JA3/JA3S?

Prompt 17

Explain this parser.py code line by line. I want to understand exactly what every offset and length calculation is doing.

Prompt 18

What are the limitations of this parser? In particular, what happens if the TLS handshake is split across multiple TCP packets?

4. Implementing JA3/JA3S

Prompt 19

{Our Code for ja3.py}

Check whether my JA3 string construction follows the official JA3 field order and whether GREASE is being removed correctly.

Prompt 20

How should I validate that my JA3 implementation is actually correct and not just producing some hash?

Prompt 21

Give me a known published JA3 example that I can use to test my implementation and tell me what output I should get.

5. Redis Database and Identification

Prompt 22

Help me design `populate_db.py` so I can capture a `ClientHello` and associate its JA3 fingerprint with a client name like `curl`, `Python` or `OpenSSL`.

Prompt 23

What should happen when the JA3 is not present in the database? Can I add fuzzy matching as a fallback?

6. Testing the Implementation

Prompt 24

Before testing live traffic, let's write unit tests for JA3. What cases should we test including GREASE and empty fields?

Prompt 25

My tests are passing, but I want to validate the implementation against actual published JA3 examples instead of only values that we created ourselves. How should I do that?

Prompt 26

Now let's test the live capture with `curl`. Give me the simplest way to run the sniffer and generate one HTTPS request so we can verify the complete pipeline.

Prompt 27

The sniffer is running but I'm not seeing `ClientHello`. What should I check first? I'm running this on Windows.

7. Testing Different Clients

Prompt 28

I need at least five different clients/tools for the project. What can I use to generate clearly different TLS ClientHello messages on my Windows machine?

Prompt 29

Help me create a custom Python TLS client where I can control the TLS version and cipher configuration so I can demonstrate that it produces a different fingerprint.

Prompt 30

How can I generate TLS traffic using curl, Python, Node.js and OpenSSL while my Scapy sniffer is running?

Prompt 31

I am getting unknown JA3 fingerprints while testing. Why is this happening when I am only generating traffic from the client I want to test?

8. Debugging and Refinement

Prompt 32

My live sniffer is capturing packets but sometimes the JA3 is unknown. Explain what could be causing this and how I should debug it.

Prompt 33

My populate_db.py currently stores the first ClientHello it sees. Is this reliable if there are other applications generating HTTPS traffic in the background?

Prompt 34

Should I modify the capture code to filter by destination IP, source port or some other flow information when registering a fingerprint?

Prompt 35

I got this Redis/Docker error when starting the project. Explain what it means and whether my stored fingerprints are lost.

Prompt 36

I am using Windows PowerShell and curl commands are behaving differently from Linux commands. What is the correct way to generate HTTPS traffic with curl.exe?

Prompt 37

Something interesting happened: the same custom client gave me two different JA3 hashes on two runs even though the code and configuration were the same. Help me investigate what changed in the actual ClientHello.

9. Validation and Final Demonstration

Prompt 38

Does my current project satisfy all the expected outcomes from the original project description? Go through every requirement one by one and tell me what is complete and what is still missing.

Prompt 39

I need to demonstrate curl vs browser vs custom TLS client purely from their handshake fingerprints. What is the cleanest experiment I can perform?

Prompt 40

I have completed the implementation and testing. Help me make a final checklist against the project description so I don't miss anything before submission.

Prompt 41

Explain my complete project architecture in a way that I can use to prepare for a project viva, including what every file does and how data flows through the system.

Prompt 42

Finally review the README and project writeup for technical accuracy. I don't want to claim things like JA4 or eBPF are implemented when they are only future work.